Bootstrapping, Measured: Toward an Operational Definition of Collective IQ in Research Teams

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Abstract

Engelbart argued that groups should improve the way they improve. We propose three operational measures of a research team's collective capability, namely time-to-reuse of prior work, cross-member retrieval success and decision traceability, and pilot them with four lab groups over eight weeks. Teams with shared, searchable records had faster reuse but not better traceability.

1 Introduction

"Collective IQ" is a powerful phrase and an unmeasured one. Without measures, claims that a tool augments a group cannot be tested.

2 Measures

• Time-to-reuse: days between a piece of work being produced and its first reuse by another member.

• Cross-member retrieval: share of successful attempts by one member to find something another member produced, in a timed task.

• Decision traceability: share of a sample of past decisions for which the team can name the evidence and the meeting where it was decided.

3 Pilot

Four lab groups (5-9 members) participated for eight weeks. Two used a shared, searchable notes system; two used individual tools.

Group	Shared records	Median time-to-reuse (days)	Retrieval success	Traceability
A	yes	6	72%	41%
B	yes	9	68%	35%
C	no	17	44%	38%
D	no	21	39%	30%

4 Discussion

Shared records helped people find things but did not help them reconstruct why decisions were made. Decisions lived in meetings and chats that no one recorded.

5 Limitations

Four groups, eight weeks, and measures that need validation.

References

[1] Engelbart, D. C. Augmenting Human Intellect: A Conceptual Framework. SRI, 1962.

[2] Hutchins, E. Cognition in the Wild. MIT Press, 1995.